ADVANCED INDUSTRIAL ORGANIZATION

LECTURE 2

Static Imperfect Competition

Price, quantity, capacity and strategic interaction

Study notes based on the lectures of Hui Shen

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Reading guide

These notes follow Lecture 2, based on Hui Shen's slides dated September 4, 2026. The section and subsection structure follows the course outline; smaller explanatory headings develop the material within that structure. Progressive builds are consolidated without omitting their additional arguments, examples or calculations.

Original **Model numbers** identify the classroom models; each is followed by a descriptive name. Course figures are redrawn with their axes and economic interpretation. Numbered takeaways separate the conclusions, their mechanisms and the assumptions that support them.

A model is read in the order

$$\text{context} \longrightarrow \text{choices} \longrightarrow \text{optimization} \longrightarrow \text{equilibrium} \longrightarrow \text{meaning}.$$

For a sequential model, start with the last decision and work backwards. For a derivative, identify the variable being changed and what remains fixed. A first-order condition is followed by the relevant feasibility, boundary and deviation checks.

Definitions and results

Blue boxes state the conclusions and the assumptions under which they hold. A displayed optimum is a conclusion about the specified model, rather than a rule for every industry.

Economic mechanism

Green boxes explain the incentives and distinguish a direct effect from an equilibrium response. In particular, a price reduction is a transfer unless it also changes quantities, matching or other real outcomes.

Examples, exercises and lecture cases

Purple boxes contain calculations or applications. Historical cases retain their original period and numerical context. An exercise's hypothetical assumptions are distinguished from historical facts.

Conditions and boundary cases

Amber boxes keep the qualifications that matter for a result: market coverage, nonnegative quantities, integer entry, credible commitment, or an approximation that neglects a firm's effect on an aggregate.

Derivations are beside the model. Boxes labelled "Lecture derivation" develop the classroom calculations and proofs, including those placed in the slides' appendix. Intermediate algebra and explanatory steps are included where needed; no separate mathematical appendix is required to follow the argument.

Instructor stars. The symbol $\star$ preserves an instructor-starred topic. It identifies the course's starred or supplementary material, not an additional importance ranking. Starred sections retain their star in the contents, and the notes state when the slides identify them as skipped.

Notation

Symbol	Meaning
q_i, $Q = \sum_i q_i$	Firm output and total industry output.
$Q_i(\boldsymbol{p})$, $P(Q)$	Direct demand and inverse market demand.
$C_i(q_i)$, $MC_i = C_i'(q_i)$	Total economic cost and marginal cost.
π_i	Profit; specify whether committed fixed costs are included.
$s_i = q_i/Q$, $H = \sum_i s_i^2$	Market share and normalized Herfindahl index.
$\eta = -Q'(p)p/Q(p) > 0$	Absolute price elasticity of demand.
τ, r_i, M	Mismatch/transport cost, valuation and consumer mass.
a, b, d	Demand intercept, own-quantity coefficient and cross-quantity coefficient.
λ, γ	Conduct parameter and conjectural variation.
$R_i(x_j)$	Best response of firm i to a rival's action.
$[z]_+ = \max\{z, 0\}$	Positive part, used for nonnegative demand or output.
CS, TS	Consumer surplus and total surplus.

Three distinctions to keep visible. A product's substitutability is a feature of demand; strategic substitutability is a feature of incentives. A fixed cost is independent of current output; a sunk cost cannot be recovered. A first-order condition is an optimization condition; equilibrium also requires that all agents' choices fit together and that their constraints are respected.

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SECTION 1

Introduction to Part II: Market Power

Lecture 2 · source PDF pp. 1–23

Oligopoly means strategic interaction

An oligopoly has few suppliers whose decisions affect one another. Firms are neither price takers nor unconstrained monopolists. The lecture's examples are soft drinks, cars, smartphones, steel, natural gas and beer; local service markets can have the same strategic structure.

A game specifies players, actions, information, timing and payoffs. In a Nash equilibrium, no firm can gain by changing its action while rivals' actions remain fixed. Anticipating that rivals optimize does not mean they change their actions during a unilateral deviation.

Case 0: the DVD-by-mail industry

1. **Sequence.** Netflix launched subscriptions in 1999 and initially faced limited competition. Wal-Mart and Blockbuster entered and competed aggressively in the early 2000s; the lecture emphasizes the 2004 price cuts and Netflix's response.
2. **Strategic surprise.** Netflix's management had underestimated the likelihood and significance of entry. Amazon was a potential entrant: anticipated competition could constrain current decisions even before entry occurred.
3. **Three connected decisions.** Subscription prices are current actions; stocking costly new-release copies is a capacity commitment; entry and subsequent price responses add timing and repetition. Expected future price wars can influence whether firms enter.

Cournot's 1838 and Bertrand's 1883 analyses specify different commitments. No single oligopoly model suits every industry. This lecture compares price and quantity choices, connects capacity to pricing, and develops strategic substitutes and complements.

The lecture's three learning objectives

1. Solve the basic price and quantity games, identifying what each firm holds fixed in a unilateral deviation.
2. Compare their predictions using capacity commitments and a common differentiated-demand system.
3. Use strategic complements and substitutes to understand reaction slopes and comparative statics.

The final, starred part studies empirical market-power measurement. The private-cost topic is also starred and marked as skipped in the slides; it is developed here. Lecture 3 adds sequential choice, endogenous entry, repeated competition and industry dynamics.

SECTION 2

Price competition

2.1 The standard Bertrand model

Lecture 2 · source PDF pp. 24–44

Model 1: Homogeneous Bertrand competition with identical costs

Two firms offer an identical good, have constant marginal cost c, face no capacity limit and simultaneously post prices. Market demand is $Q(p)$, positive near c and decreasing. The lower-priced firm serves the whole market. At a tie, firm i receives fraction α_i, with $0 < \alpha_i < 1$ and $\alpha_1 + \alpha_2 = 1$:

$$Q_i(p_i, p_j) = \begin{cases} Q(p_i), & p_i < p_j, \\ \alpha_i Q(p_i), & p_i = p_j, \\ 0, & p_i > p_j. \end{cases} \tag{2.1}$$

Profit is $(p_i - c)Q_i$. These demand functions are discontinuous at a price tie above cost.

A deviation proof

Lecture derivation: Eliminate all profitable one-firm deviations

At $p_1 = p_2 = c$, both firms earn zero operating profit. A price rise loses all demand and leaves profit zero; a price cut below cost loses money. Thus the profile is a Nash equilibrium. To establish uniqueness under the standard assumptions, eliminate other profiles:

1. If both prices are above c and unequal, the high-priced firm can undercut the lower price while remaining above c, obtain all demand and earn positive profit instead of zero.
2. If the firms tie at $p > c$ with positive demand, a sufficiently small undercut captures the whole market. The discontinuous quantity gain outweighs the arbitrarily small reduction in margin. If demand at the tied price is zero, a firm instead chooses a price above cost with positive demand and earns a positive profit.
3. If $p_i > p_j = c$, the low-priced firm can raise its price to a point strictly between c and p_i, retain all demand and make positive profit.
4. If the lowest posted price is below c and obtains positive sales, at least one selling firm makes a loss and can avoid it by raising its price to a nonselling level or to c.

Lesson 1: the Bertrand outcome

With homogeneous products, identical constant marginal costs, unrestricted capacity and simultaneous continuous prices, the unique price pair in the two-firm model is $p_1 = p_2 = c$. The equilibrium price-cost margin and operating profit are zero even with only two firms.

1. A small undercut can capture the entire market when capacity is unrestricted.
2. A price above common marginal cost therefore cannot survive the deviation check.
3. This conclusion relies on homogeneous products and the stated cost, timing and capacity assumptions; firm count alone does not determine market power.

For more than two symmetric firms, a profile with at least two firms charging c and all others charging no less than c is an equilibrium. Concentration alone therefore cannot identify market power.

Best responses and the open-set problem

Let p^M maximize $(p-c)Q(p)$ and assume monopoly profit rises on (c, p^M). Against a rival price p_j in that interval, a firm wants to undercut by an arbitrarily small amount. With a continuous price space there is no *largest* price strictly below p_j. Matching p_j loses a fraction of demand; every strict undercut can be improved by one closer to p_j. The best response need not exist.

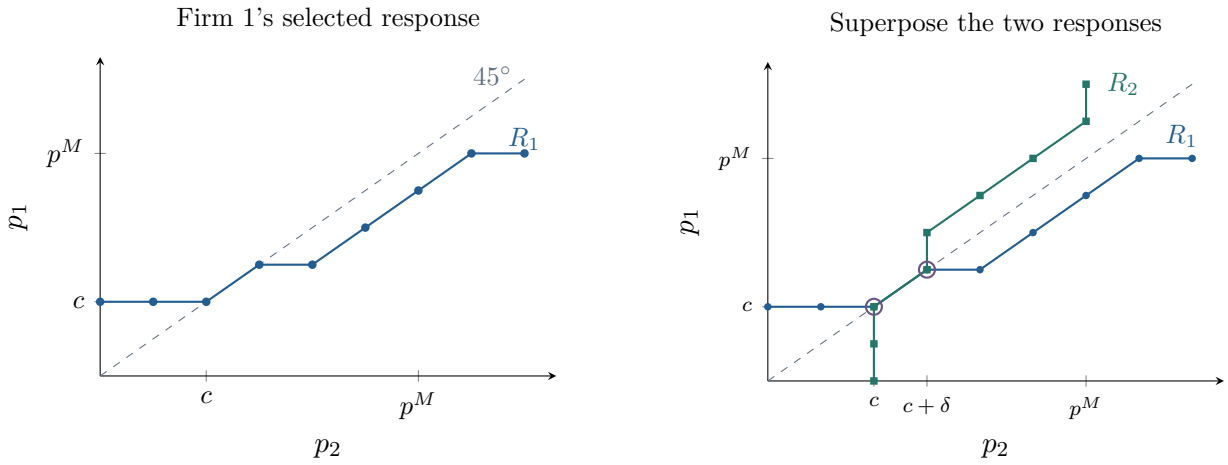
If $p_j < c$, losing all demand gives zero profit and is preferable to undercutting; $p_i = c$ is one such response. If $p_j > p^M$, charging p^M wins all demand and maximizes profit. Against $p_j = c$, any $p_i \geq c$ yields zero profit. These are best-response *correspondences*, not necessarily single-valued curves.

A discrete price grid changes the equilibrium set

On a grid with step δ and equal sharing, $p_1 = p_2 = c + \delta$ can be an equilibrium because matching earns a positive margin while a one-step undercut earns zero. A tie at $c + 2\delta$ requires

$$\delta Q(c + 2\delta) \geq \delta Q(c + \delta).$$

It fails for strictly decreasing demand, although it can hold with locally constant demand. Thus the exact claim of three grid equilibria at c, $c + \delta$ and $c + 2\delta$ is not universal. As the grid becomes fine, the usual admissible markups converge to zero.



Course diagrams: a single Bertrand response and the response pair. Source PDF pp. 32–36. The axes retain the course orientation. This exact finite-grid illustration uses $Q(p) = 10 - p$, $c = 2$, $\delta = 1$ and equal sharing; circles identify the two equilibrium ties c and $c + \delta$. Points are feasible grid choices and connecting lines guide the eye. At rival prices at or below cost, the plotted price is one zero-profit selection. In continuous prices the intermediate "just below target is a supremum, as explained above; it is not an attained response curve. The monopoly-price plateau and the diagonal comparison are retained.

The intuition of an upward-moving undercutting target is useful, but a smooth best-response diagram should not conceal the discontinuity of homogeneous Bertrand demand. Differentiated products will give well-defined smooth responses.

Asymmetric marginal costs

Model 2: Homogeneous Bertrand competition with heterogeneous costs

Let $c_1 < c_2 < \cdots < c_n$. Suppose a price tie is allocated to the more efficient firm. Let p_1^M be firm 1's unconstrained monopoly price. If both leading firms post a common $p \in [c_1, \min\{c_2, p_1^M\}]$ and other firms post no lower price, firm 1 wins the market. It does not gain from cutting an already profitable price, and the less efficient firms cannot profitably undercut below their costs. This produces a continuum of Nash outcomes supported by nonselling firms' prices.

A less efficient firm posting a price below its cost is fragile: a small probability of receiving demand would make that strategy loss-making. Requiring the inefficient firms not to price below their costs selects the familiar limit-pricing outcome

$$p_1^* = \min\{c_2, p_1^M\}, \qquad p_2^* = c_2, \tag{2.2}$$

with the efficient tie-break applying when $p_1^* = c_2$. Higher-cost firms can post prices at least as high. If $p_1^M \geq c_2$, firm 1's operating profit is

$$\pi_1 = (c_2 - c_1)Q(c_2).$$

If the cost advantage is so large that $p_1^M < c_2$, the rival does not constrain the monopoly price. Positive profit under price competition can therefore reflect an efficiency advantage rather than a failure of the undercutting logic.

A classroom extension: willingness to pay can create the same rent

The handwritten notes compare two products with common valuations $v_1 > v_2 > 0$ and zero marginal costs. Everyone prefers product 1 at equal prices. If firm 2 charges zero, firm 1 can charge $p_1 = v_1 - v_2$ and just match the rival's net utility; a tie-break toward product 1 gives it the whole unit mass of buyers. Its profit is $v_1 - v_2$ and consumer surplus is v_2. Charging more loses buyers; charging less needlessly sacrifices margin. If prices are continuous and ties do not favor firm 1, the same expression is a limiting undercutting outcome rather than automatically a pure equilibrium.

This example separates horizontal taste differences from an advantage valued identically by all consumers. It also makes the tie-breaking convention explicit.

2.2 Price competition with uncertain costs*

Model 3: Private uniform costs and a Bayesian price schedule

Instructor-starred topic; marked as skipped in the source overview.

Lecture 2 · source PDF pp. 45–68

From a known rival cost to a distribution of costs

With symmetric known costs, charging exactly c gives zero profit in every circumstance. A price above cost also yields at least zero profit and can earn positive profit if the rival posts a sufficiently high price. Thus $p = c$ can be weakly dominated without invalidating its Nash-equilibrium status. This observation motivates an information change: what if each firm knows only its own cost?

Let there be $n \geq 2$ firms. Costs c_i are independent draws from the uniform distribution on $[0, 1]$, each observed privately. Products are homogeneous, capacity is unrestricted and $Q(p) = 1 - p$ for $0 \leq p \leq 1$. A lower price wins all demand, with ties split. Independence, continuity and a strictly increasing symmetric price schedule make ties probability zero.

Seek a symmetric Bayesian Nash strategy $p(c)$ with $p'(c) > 0$. Here the strategy is a *mapping from private information to an action*, not one price common to all types.

Deriving the equilibrium price schedule

Lecture derivation: Winning probabilities, the inverse strategy and the equilibrium ODE

Let $h = p^{-1}$. If firm i posts p_i, it beats one rival with probability

$$\Pr[p_i < p(c_j)] = \Pr[h(p_i) < c_j] = 1 - h(p_i).$$

Independence makes the probability of beating all $n - 1$ rivals equal to $[1 - h(p_i)]^{n-1}$. A type c therefore maximizes

$$(p - c)(1 - p)[1 - h(p)]^{n-1}. \tag{2.3}$$

The three factors are margin, quantity conditional on winning, and probability of winning. The first-order condition is

$$(1 + c - 2p)[1 - h(p)]^{n-1} \tag{2.4}$$

$$- (p - c)(1 - p)(n - 1)[1 - h(p)]^{n-2}h'(p) = 0. \tag{2.5}$$

Lecture derivation: Winning probabilities, the inverse strategy and the equilibrium ODE (continued)

On the equilibrium path, $h(p(c)) = c$ and $h'(p(c)) = 1/p'(c)$. Cancel the common factors and rearrange:

$$p'(c) = \frac{(n-1)(p(c)-c)(1-p(c))}{(1-c)(1+c-2p(c))}. \tag{2.6}$$

Try an affine schedule $p(c) = A + Bc$. The upper-end normalization is $p(1) = 1$, so $A + B = 1$. Then

$$p(c) - c = A(1-c), \quad 1 - p(c) = B(1-c), \quad 1 + c - 2p(c) = (1-2A)(1-c).$$

Substitution into Equation (2.6) gives $B = (n-1)AB/(1-2A)$. With $B > 0$, this reduces to $1 = (n+1)A$ and $B = n/(n+1)$.

A symmetric Bayesian equilibrium

$$\boxed{p(c) = \frac{1+nc}{n+1}, \qquad p(c) - c = \frac{1-c}{n+1}.} \tag{2.7}$$

Every type $c < 1$ posts above cost. More efficient types charge lower prices and have larger absolute margins.

The type $c = 1$ earns zero with any nonloss-making price that never wins positive demand; $p(1) = 1$ is a natural monotone normalization at a probability-zero endpoint. Choosing demand intercept one aligns the upper cost support with the choke price; this is substantive normalization for this solved example, not a general reason that intercepts above one are irrelevant.

Checking deviations, including outside the price support

Lecture derivation: Global price deviations on and off the strategy support

For prices in the candidate support $[1/(n+1), 1]$,

$$1 - h(p) = \frac{(n+1)(1-p)}{n}.$$

Hence the type-c objective, apart from a positive constant, is

$$(p-c)(1-p)^n.$$

Its derivative is

$$(1-p)^{n-1}[1 + nc - (n+1)p],$$

positive below $p(c)$ and negative above it. The candidate is therefore the global optimum within the support. Below $1/(n+1)$ the firm wins with probability one; $(p-c)(1-p)$ is increasing up to that lower support boundary for $c \geq 0$, so a still lower price cannot improve upon the boundary. Prices above one generate no sales. This checks the off-support deviations as well as the first-order condition.

Prices, expected profit and the number of firms

At type c, the probability of winning is $(1-c)^{n-1}$, so expected operating profit is

$$\mathbb{E}[\pi_i \mid c_i = c] = \frac{n}{(n+1)^2}(1-c)^{n+1}. \tag{2.8}$$

It is strictly positive for $c < 1$. This is a type's expected profit *before learning rivals' costs*; in any realized market, only the lowest-cost firm sells and receives positive revenue, except for zero-probability ties.

Let $c_{(1)} = \min_i c_i$. The transaction price is $p(c_{(1)})$. At a fixed realized minimum cost,

$$p(c_{(1)}) - c_{(1)} = \frac{1 - c_{(1)}}{n+1}$$

falls as n increases: there is a competitive-pressure effect. Additionally, the expected lowest cost falls because $\mathbb{E}c_{(1)} = 1/(n+1)$. Thus

$$\mathbb{E}p_{\text{transaction}} = \frac{2n+1}{(n+1)^2}, \qquad \mathbb{E}Q = 1 - \frac{2n+1}{(n+1)^2}. \tag{2.9}$$

Both selection toward an efficient seller and stronger competition reduce expected price. As $n \to \infty$, the transaction price and minimum cost approach zero, output approaches one, and margins vanish.

Lesson 2: private costs soften homogeneous price competition

1. Every type $c < 1$ posts above cost and earns positive expected operating profit before learning rivals' costs.
2. More firms lower the margin at a given cost and lower the expected minimum cost; both channels lower the transaction price and increase total output, while each cost type extquotesingle s expected operating profit falls.
3. Only the lowest-cost firm sells in a realized market. Expected profit across rival draws is distinct from revenue in one realization; competition reaches its limit as n grows.

Why a high posted price can still be rational

A costly firm loses in most realizations, but it can sell when all rivals are still more costly. If costs fluctuate over time, the identity of the winning seller changes. The lecture's online-marketplace example, such as Taobao listings that remain available when cheaper supply is absent, illustrates this logic. Persistent simultaneous sales by many differentiated sellers require additional ingredients.

The model gives positive expected operating profits without fixed entry costs. It does not explain free entry or guarantee net profitability after an entry payment.

2.3 Price competition with differentiated products

Lecture 2 · source PDF pp. 69–105; 287–289

Why differentiation relaxes the undercutting pressure

If products are imperfect substitutes, a small price increase need not lose every customer. Perceived differentiation can exist even with the same chemical composition, as with branded and generic drugs; the lecture uses branded Aspirin and a chemically equivalent unbranded pill as an example. It can reflect location, taste, quality, reliability, information or brand perceptions.

Case 1: bananas, oranges and perceived differentiation

1. **Question.** Even a fruit market can contain differentiated products. The lecture uses Sunkist citrus and Chiquita bananas.
2. **Mechanism.** California citrus growers created the Sunkist trademark more than a century ago, allowing advertising without confusion with competitors. The brands convey expectations about taste, freshness or consistency; Chiquita bananas also sell at a premium.
3. **Lesson.** Demand depends on perceived attributes and reliability. A brand premium alone does not establish a physical quality difference; the preceding Aspirin/generic example makes the same distinction.

Symmetric Hotelling price competition

Model 4: Endpoint Hotelling price competition

Two firms locate at the endpoints $l_1 = 0$ and $l_2 = 1$ of a unit interval. A unit mass of consumers is uniformly distributed along it and each buys at most one unit. Both firms have marginal cost c. Consumer x receives

$$v_i(x) = r - \tau|x - l_i| - p_i,$$

or zero from abstaining. Begin with a covered market and positive demand for both firms.

Lecture derivation: Find the marginal buyer, then optimize each price

Indifference gives

$$\hat{x} = \frac{1}{2} + \frac{p_2 - p_1}{2\tau}, \qquad Q_i = \frac{1}{2} + \frac{p_j - p_i}{2\tau}.$$

Firm i maximizes

$$\pi_i = (p_i - c)\left(\frac{1}{2} + \frac{p_j - p_i}{2\tau}\right).$$

Differentiate, keeping p_j fixed:

$$\frac{\partial \pi_i}{\partial p_i} = \frac{1}{2} + \frac{p_j - p_i}{2\tau} - \frac{p_i - c}{2\tau} \tag{2.10}$$

$$= \frac{p_j - 2p_i + c + \tau}{2\tau} = 0. \tag{2.11}$$

The second derivative is $-1/\tau < 0$ in this region. The best response is

$$R_i(p_j) = \frac{p_j + c + \tau}{2}. \tag{2.12}$$

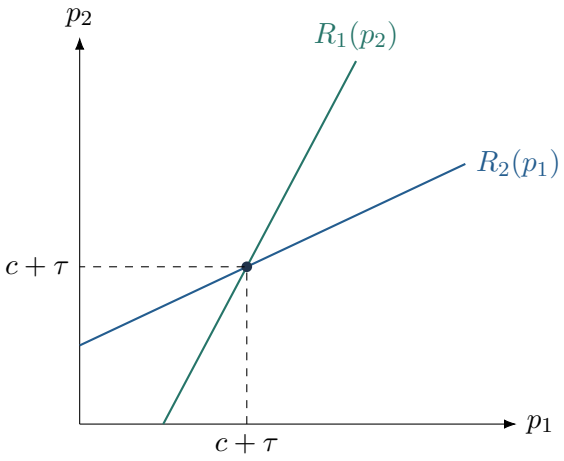
It slopes upward: a higher rival price raises the optimal own price. Solving the two simultaneous equations gives

Lesson 3: differentiation creates market power

$$p_1^* = p_2^* = c + \tau, \quad Q_1^* = Q_2^* = \frac{1}{2}, \quad \pi_i^* = \frac{\tau}{2}. \tag{2.13}$$

At these prices, full coverage requires $r \geq c + 3\tau/2$. The marginal consumer at $x = 1/2$ has the lowest utility, $r - c - 3\tau/2$.

1. Differentiation makes a small price increase lose some customers rather than the whole market.
2. Raising τ raises the equilibrium markup in this covered regime.
3. Participation must be checked: once abstention matters, the relevant demand branch and price formula can change.



A larger τ raises the markup because traveling to the rival is more costly. The locations are fixed in this price game; choosing locations would be a different game.

The outside option changes the regime

The covered-market formula cannot be extended indefinitely as τ rises. When τ is sufficiently high, each firm serves a local group and the rivals' markets no longer meet. Then local demand is $(r - p_i)/\tau$ and

$$\max_{p_i}(p_i - c)\frac{r - p_i}{\tau} \quad \Longrightarrow \quad p_i^{local} = \frac{r + c}{2}.$$

The local market radius is $(r - c)/(2\tau)$, so disjoint markets occur when $\tau > r - c$. Between the fully competitive covered regime and disjoint local monopolies, participation can bind at the center.

A complete symmetric regime check

Let $A = r - c > 0$. For the endpoint model with outside utility zero, the symmetric price regimes are

$$p^* = \begin{cases} c + \tau, & \tau \le 2A/3, \\ r - \tau/2, & 2A/3 \le \tau \le A, \\ (r + c)/2, & \tau \ge A. \end{cases}$$

In the middle region, the marginal consumer's utility is zero and demand is kinked; the interior covered-demand first-order condition no longer applies. In the last region, the market is only partly covered. Differentiation affects both substitution and participation.

Case 2: Airbus, Boeing and capacity constraints

1. **Historical setting.** The lecture presents Airbus and Boeing as a wide-bodied-aircraft duopoly, with a forecast of approximately US$2.6 trillion over two decades.
2. **Entry barriers.** Developing a new type can cost up to US$10 billion. Safety and reliability records take decades to establish, constraining potential Chinese and Russian entrants.
3. **Why capacity matters.** The A380 was about two years late; the 787 faced delays in 2007–2008. A supplier cannot instantly serve every buyer at its posted price. Capacity commitments and delivery time therefore motivate the later two-stage model. The figures are historical teaching material.

A circular extension: the Salop model

Model 5: Salop pricing with localized competition

Place $n \ge 3$ firms at equal distances $1/n$ on a circle of circumference one. A unit mass of consumers is uniformly distributed around the circle; marginal cost is c, mismatch cost per unit distance is τ, and valuations are high enough for coverage. Near a symmetric equilibrium, a firm competes with its two neighbors.

Lecture derivation: Two neighbouring cutoffs determine Salop demand

On an unwrapped local segment, the indifferent consumer between i and $i+1$ is

$$\hat{x}_{i,i+1} = \frac{2i+1}{2n} + \frac{p_{i+1} - p_i}{2\tau}.$$

The boundary on the left is

$$\hat{x}_{i-1,i} = \frac{2i-1}{2n} + \frac{p_i - p_{i-1}}{2\tau}.$$

Subtracting them,

$$Q_i = \frac{1}{n} + \frac{p_{i-1} + p_{i+1} - 2p_i}{2\tau}. \tag{2.14}$$

If both neighbors charge p, firm i chooses p_i to maximize $(p_i - c)[1/n + (p - p_i)/\tau]$. Its first-order condition is

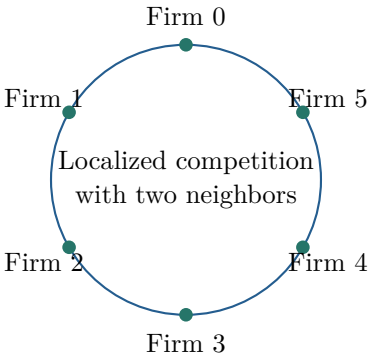
$$\frac{1}{n} + \frac{p - 2p_i + c}{\tau} = 0.$$

Lecture derivation: Two neighbouring cutoffs determine Salop demand (continued)

Setting $p_i = p$ gives

$$p^* = c + \frac{\tau}{n}, \qquad Q_i^* = \frac{1}{n}, \qquad \pi_i^* = \frac{\tau}{n^2}. \tag{2.15}$$

The farthest consumers in a firm's segment are $1/(2n)$ away, so $r \geq c + 3\tau/(2n)$ is the associated coverage condition. More firms create closer substitutes and reduce the markup; as $n \to \infty$, price approaches marginal cost.



2.4 Asymmetric competition with differentiated products

Lecture 2 · source PDF pp. 96–105, 287–289

Model 6: Hotelling pricing with a common quality advantage

Return to the line and let valuations differ: $v_i(x) = r_i - \tau|x - l_i| - p_i$, with $\Delta = r_1 - r_2 > 0$. Assume $\Delta < \tau$, as in the lecture, so some consumers intrinsically prefer product 2 despite its lower baseline quality. Also maintain full coverage; at the equilibrium derived below this requires $(r_1 + r_2)/2 \geq c + 3\tau/2$.

Lecture derivation: Solve the asymmetric price system and recover market shares

The indifferent consumer first solves

$$r_1 - \tau\widehat{x} - p_1 = r_2 - \tau(1 - \widehat{x}) - p_2.$$

Rearranging gives $2\tau\widehat{x} = \tau + \Delta - p_1 + p_2$. Consumers to the left buy product 1, so demand is

$$Q_1 = \frac{1}{2} + \frac{\Delta - (p_1 - p_2)}{2\tau}, \qquad Q_2 = 1 - Q_1.$$

With common cost c, the first-order conditions are

$$2p_1 - p_2 = c + \tau + \Delta, \qquad 2p_2 - p_1 = c + \tau - \Delta.$$

Solving,

$$p_1^* = c + \tau + \frac{\Delta}{3}, \qquad p_2^* = c + \tau - \frac{\Delta}{3}, \tag{2.16}$$

$$Q_1^* = \frac{1}{2} + \frac{\Delta}{6\tau}, \qquad Q_2^* = \frac{1}{2} - \frac{\Delta}{6\tau}. \tag{2.17}$$

The high-quality firm charges more and sells more. Its advantage is partly absorbed by a price premium, so the difference in market shares is smaller than it would be at equal prices.

The planner's allocation

Lecture derivation: Prices cancel from covered-market surplus; matching still matters

Let z be the cutoff allocating consumers on $[0, z]$ to firm 1 and the remainder to firm 2. Each transaction with firm i contributes valuation minus production cost minus mismatch cost. Price cancels between consumer and producer. Therefore

$$\mathrm{TS}(z) = \int_0^z (r_1 - c - \tau x)\,\mathrm{d}x + \int_z^1 (r_2 - c - \tau(1-x))\,\mathrm{d}x \tag{2.18}$$

$$= -\tau z^2 + (\Delta + \tau)z + r_2 - c - \frac{\tau}{2}. \tag{2.19}$$

Differentiate:

$$\mathrm{TS}'(z) = \Delta + \tau - 2\tau z = 0, \qquad \mathrm{TS}''(z) = -2\tau < 0.$$

The efficient interior cutoff and equilibrium cutoff are

$$z^{eff} = \frac{1}{2} + \frac{\Delta}{2\tau}, \qquad z^{NE} = \frac{1}{2} + \frac{\Delta}{6\tau}. \tag{2.20}$$

Thus $z^{NE} < z^{eff}$: the better product serves too few consumers from a total-surplus perspective. Under full coverage, equal prices implement the efficient cutoff. The choice $p_1 = p_2 = c$ does so if valuations also support participation; equal prices at an arbitrarily high level need not cover the market. The welfare loss from the distorted cutoff is

$$\mathrm{TS}(z^{eff}) - \mathrm{TS}(z^{NE}) = \tau(z^{NE} - z^{eff})^2 = \frac{\Delta^2}{9\tau}.$$

Lesson 4: an efficient firm can be too small

In this model, the high-quality firm sells too little because it uses some of its advantage to charge a higher price. With equal valuations and $c_1 < c_2$, the analogous advantage is $c_2 - c_1$: the efficient cutoff shifts by $(c_2 - c_1)/(2\tau)$, but the equilibrium cutoff shifts by only $(c_2 - c_1)/(6\tau)$, when both remain interior.

There are three implications:

1. Higher quality increases both the equilibrium price and the equilibrium share, but the premium prevents the better product from serving all consumers for whom it is socially preferable.
2. Under full coverage, equal prices implement the efficient cutoff; marginal-cost pricing also needs participation to be feasible.
3. A subsidy to the better firm or a tax on the worse firm can correct the relative incentive in this model.

The policy conclusion It is model-dependent: participation, the funding of subsidies, entry and other distortions must be specified before making a general policy recommendation. It does explain why protecting a weak firm is not automatically an efficiency improvement.

SECTION 3

Quantity competition

Lecture 2 · source PDF pp. 106–150

What does a quantity commitment mean?

Each firm chooses an output $q_i \geq 0$ simultaneously. The market price clears total output, $p = P(Q)$, where $Q = \sum_i q_i$. A firm treats its rivals' *quantities* as fixed when changing its own quantity.

An auctioneer who sells a precommitted supply is one literal interpretation. Capacity committed before flexible prices are chosen is another, developed in the next chapter. Seeing actual firms announce prices does not alone establish that homogeneous Bertrand is the correct model; ask what is difficult to adjust.

3.1 The linear Cournot model

Model 7: Linear Cournot competition with heterogeneous costs

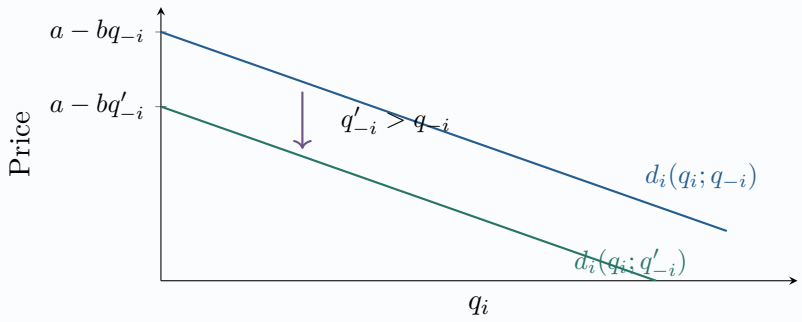
Residual demand and the individual choice problem

Let $P(Q) = a - bQ$, $a, b > 0$, and let firm i have cost $C_i(q_i) = c_i q_i$, with $0 \leq c_i < a$. Write $Q_{-i} = \sum_{j \neq i} q_j$. Firm i's residual inverse demand is

$$p = (a - bQ_{-i}) - bq_i.$$

A larger rival output lowers the intercept but leaves the slope unchanged. Conditional on Q_{-i}, firm i acts as a monopolist on this residual demand.

Lecture derivation: Own-output optimization with rivals held fixed



Course figure: residual inverse demand in Cournot. Source PDF p. 115. Holding rivals' output fixed gives $d_i(q_i; q_{-i}) = a - bq_{-i} - bq_i$. More rival output shifts this line down without changing its slope. The firm chooses its own monopoly quantity on the residual curve; it does not take the residual price as fixed.

Lecture derivation: Own-output optimization with rivals held fixed (continued)

Its objective and first derivative are

$$\pi_i = (a - bq_i - bQ_{-i} - c_i)q_i, \tag{3.1}$$

$$\frac{\partial \pi_i}{\partial q_i} = a - c_i - 2bq_i - bQ_{-i}. \tag{3.2}$$

Since the second derivative is $-2b < 0$, its constrained best response is

$$R_i(Q_{-i}) = \max\left\{0, \frac{a - c_i - bQ_{-i}}{2b}\right\}. \tag{3.3}$$

On the positive-output segment, the slope with respect to aggregate rival quantity is $-1/2$. Quantities are strategic substitutes in this model.

Solving the system rather than guessing symmetry

Lecture derivation: Sum the first-order equations, then recover each quantity

First suppose all n firms are active. The first-order condition can be written

$$a - c_i - bQ - bq_i = 0.$$

Summing over firms gives

$$na - \sum_i c_i - bnQ - bQ = 0.$$

Thus

$$Q^* = \frac{na - \sum_i c_i}{b(n+1)}, \qquad p^* = a - bQ^* = \frac{a + \sum_i c_i}{n+1}. \tag{3.4}$$

Use $bq_i = p^* - c_i$ to recover firm-level output. Let $S_{-i} = \sum_{j \neq i} c_j$; then

$$q_i^* = \frac{a - nc_i + S_{-i}}{b(n+1)}, \qquad \pi_i^* = b(q_i^*)^2 = \frac{(a - nc_i + S_{-i})^2}{b(n+1)^2}. \tag{3.5}$$

This immediately shows, within a fixed active set,

$$\frac{\partial q_i^*}{\partial c_i} = -\frac{n}{b(n+1)}, \qquad \frac{\partial q_i^*}{\partial c_j} = \frac{1}{b(n+1)} \quad (j \neq i).$$

A more efficient firm produces more and earns more. Higher rival costs relax competition and raise its output and profit. Because $p^* - c_i = bq_i^*$, the profit ranking follows directly from the output ranking.

Lesson 5: relative efficiency raises Cournot profit

1. Holding the active set fixed, a lower own cost raises own equilibrium output and profit.
2. Higher rival costs relax competition and likewise raise own output and profit.
3. The comparison applies while the activity inequalities hold. If a firm becomes inactive, solve the new active set before comparing profits.

An interior formula must pass an activity check

All n quantities are strictly positive only if

$$c_i < \frac{a + S_{-i}}{n} \quad \text{for every } i.$$

Equality gives zero output, a boundary rather than a strictly interior equilibrium. If a candidate quantity is negative, do not square it and report a positive profit. Recompute using the active firms, and verify that every excluded firm prefers zero output.

Order costs $c_1 \le \cdots \le c_n$. For a candidate active set of the first k firms, the implied price is

$$p_k = \frac{a + \sum_{i=1}^{k} c_i}{k+1}. \tag{3.6}$$

The correct active set satisfies $c_k < p_k$ and, if $k < n$, $c_{k+1} \ge p_k$. Active quantities are $(p_k - c_i)/b$; inactive quantities are zero. This implements the nonnegativity condition rather than discarding it.

Duopoly: equations and a diagram

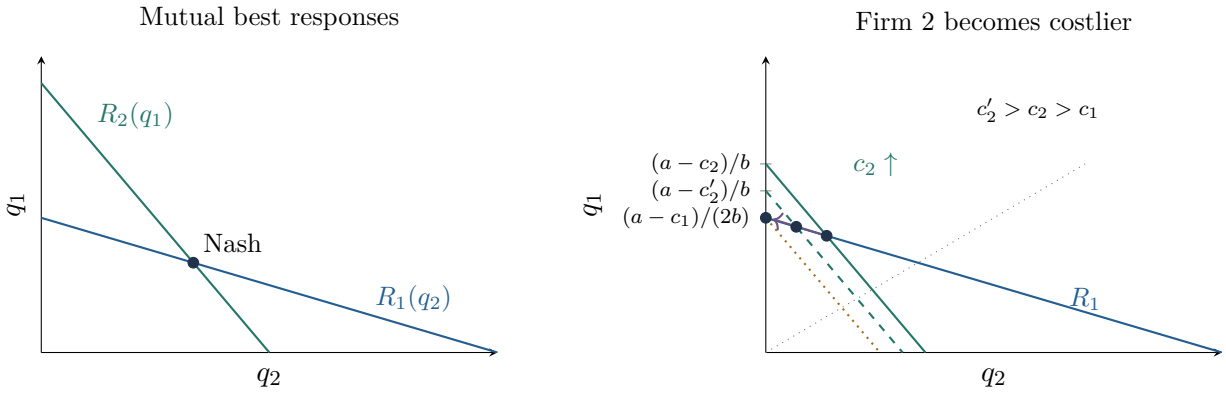
For two active firms,

$$q_1 = \frac{a - c_1 - bq_2}{2b}, \qquad q_2 = \frac{a - c_2 - bq_1}{2b}.$$

Solving the two equations yields

$$q_1^* = \frac{a - 2c_1 + c_2}{3b}, \quad q_2^* = \frac{a - 2c_2 + c_1}{3b}, \quad p^* = \frac{a + c_1 + c_2}{3}. \tag{3.7}$$

If $c_1 \le c_2$, the binding activity condition is $c_2 < (a + c_1)/2$. If it fails, firm 2 is inactive and firm 1 produces its monopoly quantity $(a - c_1)/(2b)$.



Course diagrams: Cournot intersection and a cost shift. Source PDF pp. 117 and 130–133. The first panel shows the two downward responses; the second retains the course's heterogeneous-cost shift and activity boundary. Axes follow the course: q_2 horizontal, q_1 vertical. In the second panel $a = 12, b = 1, c_1 = 2$: raising c_2 from 5 to 6 moves (q_2, q_1) from $(4/3, 13/3)$ to $(2/3, 14/3)$; at $c_2 = 7$, the boundary is $(0, 5)$. Beyond that point the two-active-firm formula is infeasible. The dotted diagonal compares the two quantities.

The course figure puts q_2 on the horizontal axis and q_1 on the vertical axis. Firm 1's response has graphical slope $-1/2$; firm 2's response is written as $q_1 = (a - c_2)/b - 2q_2$ and therefore

has graphical slope -2. A cost increase for firm 2 moves the intersection upwards along firm 1's response.

Worked heterogeneous duopoly

Let $a = 12$, $b = 1$, $c_1 = 2$ and $c_2 = 5$. Then $q_1 = 13/3$, $q_2 = 4/3$, $Q = 17/3$ and $p = 19/3$. The profits are $169/9$ and $16/9$. The activity test is $5 < (12 + 2)/2 = 7$, so both firms are indeed active. If c_2 rises to 8, the two-active-firm formula is invalid: firm 2 exits production, firm 1 supplies 5, and price is 7.

Symmetric oligopoly and the competitive limit

Set $c_i = c$ for all firms and $a > c$. Substitution into the heterogeneous formulas gives

Lesson 6: Cournot approaches the competitive allocation

$$q_i^*(n) = \frac{a-c}{b(n+1)}, \qquad Q^*(n) = \frac{n(a-c)}{b(n+1)}, \tag{3.8}$$

$$p^*(n) = \frac{a+nc}{n+1}, \qquad \pi_i^*(n) = \frac{(a-c)^2}{b(n+1)^2}, \tag{3.9}$$

$$p^* - c = \frac{a-c}{n+1}, \qquad L(n) = \frac{a-c}{a+nc}. \tag{3.10}$$

More firms reduce each firm's output and profit but increase total output. Price and the absolute margin decrease. As $n \to \infty$, $Q \to (a-c)/b$ and $p \to c$.

The zero-cost exception for a relative markup

If $c > 0$, $L(n) \to 0$. If $c = 0$, however, $L(n) = 1$ for every finite n, even though price and the absolute price-cost margin converge to zero. The relative markup divides by a price that also vanishes; the allocation still reaches the competitive limit. Distinguish a level difference from a percentage.

The comparative statics have three parts:

1. Each firm produces less and earns less as the firm count rises.
2. Total output rises, while price and the absolute price–cost margin fall.
3. The allocation approaches $p = c$ and $Q = (a-c)/b$; a relative Lerner index also vanishes when $c > 0$. The zero-cost qualification above matters for that ratio.

The mechanism is that each firm's price effect becomes small when its share of total output is small. These comparative statics condition on a given number of firms; explaining the number that enter also requires fixed or sunk entry costs.

Worked classroom extension: change the cost curvature

Lecture derivation: Quadratic own costs require new first-order conditions

The handwritten Cournot notes emphasize solving a modified cost problem before interpreting comparative statics. Consider the related exercise $P(Q) = a - Q$, $C_1(q_1) = \kappa q_1^2$ and $C_2(q_2) = c_2 q_2$, with $\kappa \geq 0$. Firm 1's cost is quadratic while firm 2's marginal cost is constant. Their interior first-order conditions are

$$a - (2 + 2\kappa)q_1 - q_2 = 0, \qquad a - c_2 - q_1 - 2q_2 = 0.$$

Use the second equation to substitute $q_2 = (a - c_2 - q_1)/2$ into the first:

$$2a - (4 + 4\kappa)q_1 - a + c_2 + q_1 = 0.$$

Thus

$$\begin{aligned} q_1^* &= \frac{a + c_2}{3 + 4\kappa}, \\ q_2^* &= \frac{a(1 + 2\kappa) - 2(1 + \kappa)c_2}{3 + 4\kappa}, \\ p^* &= \frac{(1 + 2\kappa)(a + c_2)}{3 + 4\kappa}. \end{aligned}$$

The formula requires $q_2^* > 0$. If it fails, solve firm 1's monopoly problem instead. Profits and welfare follow from their definitions:

$$\pi_1 = (1 + \kappa)(q_1^*)^2, \qquad \pi_2 = (q_2^*)^2, \qquad \mathrm{CS} = \frac{1}{2}(q_1^* + q_2^*)^2, \qquad \mathrm{TS} = \mathrm{CS} + \pi_1 + \pi_2.$$

This shows why substituting a linear-cost formula after changing the cost curvature is invalid. The method remains the same: derive own marginal cost, solve the simultaneous conditions, check activity, and only then evaluate welfare.

3.2 Implications of Cournot competition

Model 8: Cournot competition with general demand and costs

The share–elasticity markup formula

Lecture derivation: Derive the markup from the quantity first-order condition

Let $\pi_i = P(Q)q_i - C_i(q_i)$. At an interior optimum,

$$P(Q) + P'(Q)q_i = C_i'(q_i). \tag{3.11}$$

Define $s_i = q_i/Q$ and absolute market elasticity $\eta = -P(Q)/[P'(Q)Q]$. Rearranging gives

$$\boxed{\frac{P(Q) - C_i'(q_i)}{P(Q)} = \frac{s_i}{\eta}.} \tag{3.12}$$

Lesson 7: a larger share implies a larger Cournot markup

1. The interior first-order condition gives $L_i = s_i/\eta$: compared with monopoly, the price effect is scaled by the firm's share.
2. At a fixed market elasticity, a larger firm has a larger marginal markup; at a fixed share, less elastic demand allows a larger markup.
3. This is a testable restriction of homogeneous Cournot behaviour, not an unrestricted statement about every differentiated industry.

The own second derivative and the cross-partial are

$$\frac{\partial^2 \pi_i}{\partial q_i^2} = 2P'(Q) + q_i P''(Q) - C_i''(q_i), \tag{3.13}$$

$$\frac{\partial^2 \pi_i}{\partial q_i \partial q_j} = P'(Q) + q_i P''(Q), \quad j \neq i. \tag{3.14}$$

If $P' < 0$, $C_i'' \geq 0$ and $P' + q_i P'' \leq 0$, profits are strictly concave in own output. With continuous payoffs and suitably bounded convex strategy sets, standard existence arguments then apply. The nonpositive cross-partial gives strategic substitutability and downward-sloping interior best responses. Concavity alone does not establish uniqueness, and curvature conditions do not replace the boundedness or existence assumptions.

Concentration and a share-weighted markup

For general differentiable homogeneous Cournot demand, multiply Equation (3.12) by s_i and sum:

$$\bar{L} \equiv \sum_i s_i \frac{p - MC_i}{p} = \frac{\sum_i s_i^2}{\eta} = \frac{H}{\eta}. \tag{3.15}$$

With constant marginal costs c_i, the same formula is $(p - \sum_i s_i c_i)/p = H/\eta$. If there are no additional fixed costs, it implies $\sum_i \pi_i = pQH/\eta$. With nonconstant costs, the share-weighted *marginal* markup need not equal the industry's profit-to-revenue ratio.

Lesson 8: concentration measures the average Cournot markup

At a given market-demand elasticity, concentration increases the share-weighted Cournot markup. Across markets, elasticity can differ, and firms may compete in prices or face differentiated demand. Computing H alone does not establish conduct or recover a causal effect of concentration.

SECTION 4

Price vs. quantity competition

Lecture 2 · source PDF pp. 151–241

Start with the homogeneous benchmark

For homogeneous demand $Q(p) = a - p$ and common marginal cost $c < a$, two unrestricted Bertrand firms charge c, each sells $(a-c)/2$ under equal sharing, and each earns zero operating profit. Two Cournot firms instead produce $(a-c)/3$ each, charge $(a+2c)/3$ and earn $(a-c)^2/9$ each. Quantities are lower and prices and profits are higher under Cournot in this benchmark.

Lesson 9: homogeneous quantity commitment softens competition

1. At the same a and c, Cournot has higher price and lower total quantity than unrestricted homogeneous Bertrand.
2. Cournot firms receive $(a-c)^2/9$ each; homogeneous Bertrand operating profit is zero.
3. The result compares different strategic commitments. Capacity and differentiated products reveal which assumptions produce the ranking.

4.1 Limited capacity and price competition

The standard Bertrand result assumes that a tiny price cut can attract *and serve* all demand. A capacity-constrained firm may attract additional customers without being able to supply them. Edgeworth's critique makes this constraint central.

Case 3: when capacity choices condition pricing decisions

1. **Commit first.** DVD rental firms must stock costly new-release copies before serving demand. New releases attract more demand than titles released six months or a year earlier. Interfaces and pricing can steer subscribers toward back-catalogue titles.
2. **A fixed stock changes pricing.** Traditional clothing orders are placed before the season, although flexible, often vertically integrated supply chains reduce rigidity. Seats and airport slots likewise constrain festival travel; hotel rooms and rental cars are related applications.
3. **Revenue management.** When capacity is fixed and demand fluctuates, firms adapt prices and availability to sell the appropriate product to the appropriate customer at the appropriate time. This links timing, stock and price discrimination; the single-price model below isolates the capacity-commitment channel.

A two-stage game and the meaning of sunk cost

Model 9: Observable capacity commitments followed by price competition

Two firms first choose observable capacities $k_1, k_2 \geq 0$, paying ck_i in stage 1. Then they simultaneously post prices. Production up to installed capacity has zero additional marginal cost in stage 2; sales cannot exceed k_i. Demand is $Q(p) = a - p$ on its positive range.

We solve backward. A subgame-perfect equilibrium specifies an equilibrium price continuation for *every* capacity pair, including pairs not chosen on the equilibrium path. Stage-1 choices account for the price outcomes they induce.

Do not charge for capacity twice

For a price-stage sale quantity y_i, total profit is $p_i y_i - ck_i$. The payment ck_i is already sunk at that stage, so the firm maximizes revenue with respect to price. Writing $(p_i - c)y_i$ is generally wrong unless every unit of capacity is sold, $y_i = k_i$, in which case the two expressions happen to coincide.

Installed factory sizes can make capacity observable. Retail inventories may be less transparent. A local farmers' market is the lecture's example where a vendor can often see the rivals' stock. Observability is an assumption needed to interpret the chosen timing.

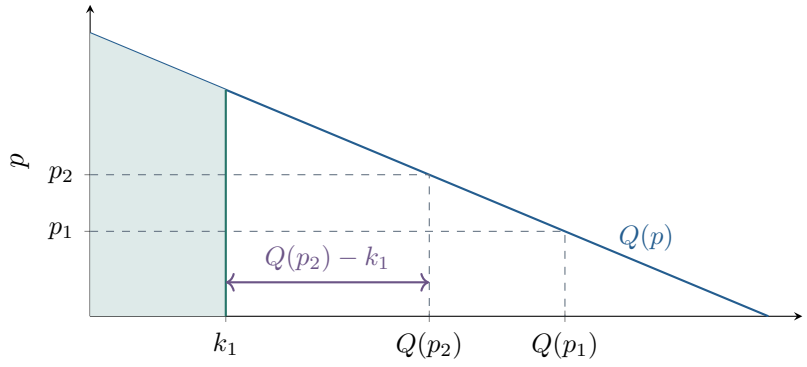
Who is served first? Efficient rationing

If a low-price firm lacks enough units to meet demand, some buyers must turn elsewhere. Under *efficient rationing*, its available units go first to consumers with the highest willingness to pay. Then if firm 1 charges less than firm 2 and has capacity k_1, firm 2's residual demand is

$$D_2^E(p_2) = \max\{a - p_2 - k_1, 0\}. \tag{4.1}$$

Actual sales are the smaller of that demand and k_2. This rule gives the high-price firm the smallest residual demand among allocations that give the low-price units to willing buyers.

A queue can rationalize this ordering only with assumptions making higher-value buyers willing and able to arrive first; willingness to pay alone does not establish a queue order. A frictionless resale market gives another rationale: low-value recipients sell to higher-value consumers. The efficient allocation serves the highest-value consumers at any given total quantity.



Course figure: efficient rationing and residual demand. Source PDF pp. 174–176. At $p_1 < p_2$, the first k_1 units go to the highest-value buyers. Among consumers willing to pay p_2, $Q(p_2) - k_1$ remain unserved by firm 1. The larger $Q(p_1)$ shows the total number who would like the low-price product. The illustration uses $a = 10, k_1 = 2, p_1 = 3, p_2 = 5$.

At equal prices, assume the allocation respects capacities and serves all feasible demand; when total demand equals $k_1 + k_2$, both capacities are fully sold.

A sufficient region for a pure price equilibrium

Lecture derivation: Check both price cuts and price increases

Given capacities, consider the common market-clearing price

$$p^* = a - k_1 - k_2 > 0. \tag{4.2}$$

At this price each firm sells its entire capacity. Test a deviation by firm 2, keeping $p_1 = p^*$.

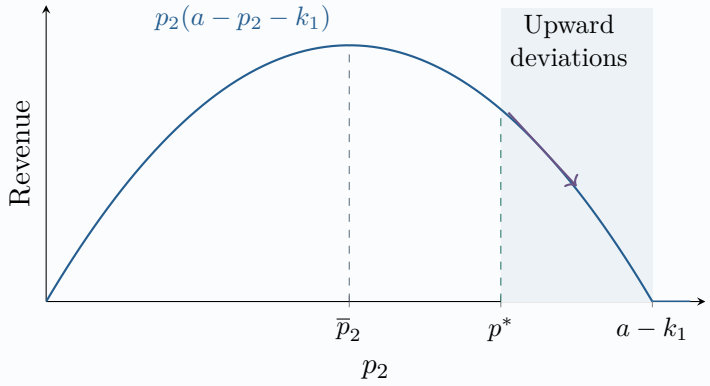
A price cut cannot increase sales. If $p_2 < p^*$, firm 2 still sells at most k_2, which it was already selling. It earns less per unit and gains nothing in volume.

A price increase gives residual demand. If $p_2 > p^*$, efficient rationing leaves $a - p_2 - k_1 < k_2$ units of residual demand, when positive. Its relevant revenue is

$$R_2(p_2) = p_2(a - p_2 - k_1).$$

This concave quadratic is maximized at

$$\bar{p}_2 = \frac{a - k_1}{2}.$$



Course figure: the revenue maximum and the proposed price. Source PDF pp. 185–186 and 190. The residual-revenue parabola peaks at $\bar{p}_2 = (a - k_1)/2$. The proposed $p^* = a - k_1 - k_2$ lies on its decreasing branch when $k_1 + 2k_2 \leq a$, so a higher price reduces revenue. The parabola ignores own capacity; only its shaded branch is used for upward deviations. Shading identifies that price region, not an additional payoff. Here $a = 10, k_1 = k_2 = 2$, giving $\bar{p}_2 = 4$ and $p^* = 6$.

A price above p^* cannot raise revenue if $p^* \geq \bar{p}_2$, or equivalently

$$k_1 + 2k_2 \leq a.$$

The analogous condition for firm 1 is $2k_1 + k_2 \leq a$.

A market-clearing price continuation

Under efficient rationing and positive capacities, the common price $p^* = a - k_1 - k_2$ is a pure Nash equilibrium of the price subgame if

$$2k_1 + k_2 \leq a, \qquad k_1 + 2k_2 \leq a. \tag{4.3}$$

Each firm's capacity is no larger than its zero-production-cost Cournot best response to the other's capacity. A profitable upward deviation is the key obstacle outside this region.

The lecture's convenient parameter restriction

Lecture derivation: Why the parameter restriction ensures small capacities

No firm that can choose zero capacity would install capacity whose cost exceeds the largest possible monopoly revenue. Since

$$\max_Q Q(a - Q) = \frac{a^2}{4},$$

a necessary bound for a nonnegative net-profit choice is

$$k_i \leq \frac{a^2}{4c}. \tag{4.4}$$

The lecture imposes

$$c < a < \frac{4}{3}c. \tag{4.5}$$

For capacities satisfying the bound,

$$k_1 + 2k_2 \leq \frac{3a^2}{4c} < a,$$

and similarly for $2k_1 + k_2$. Equivalently, the bound is below $a/3$. Thus the restriction keeps all such candidate capacities in the simple pure-price region. It is a *sufficient device for the elementary proof*, not a general economic restriction defining Cournot competition.

Stage 1 reproduces the Cournot objective

Lecture derivation: Recover the first-stage capacity equilibrium

Substitute the market-clearing continuation into full profit:

$$\tilde{\pi}_i(k_i, k_j) = (a - k_i - k_j)k_i - ck_i. \tag{4.6}$$

Differentiate with respect to k_i:

$$a - c - 2k_i - k_j = 0.$$

The stage-1 best responses and their intersection give

$$k_1^* = k_2^* = \frac{a-c}{3}, \quad p^* = \frac{a+2c}{3}, \quad \pi_i^* = \frac{(a-c)^2}{9}. \tag{4.7}$$

These are exactly the homogeneous Cournot quantities, price and profits. Capacities are chosen first, but all installed capacity is sold at the resulting price.

Lesson 10: capacity-then-price can implement Cournot

1. Under the stated efficient-rationing continuation, each firm commits to $k_i = (a-c)/3$ and sells all installed capacity.
2. The first-stage net-profit function is the Cournot quantity objective; commitment removes the benefit of a later price cut that cannot expand supply.
3. The elementary derivation uses the sufficient parameter region. Unrestricted subgame perfection also requires off-path price continuations; the broader result can involve mixed pricing.

Why price-setting firms can have Cournot outcomes

The scarce object committed first is capacity. Cutting price later cannot expand supply, so the gain from stealing the whole market disappears. The strategic first-stage payoff then has the same form as choosing an irreversible quantity directly.

A fully unrestricted subgame-perfect analysis must also specify continuations after off-path capacity choices, including very large choices. The lecture's elementary argument isolates a region with simple pure continuations. The broader capacity-then-price result under efficient rationing, associated with Kreps and Scheinkman, handles mixed-price subgames when necessary; merely substituting one on-path price is not a proof about every subgame.

Larger capacities and mixed price strategies

If $k_1 + 2k_2 > a$, firm 2 can profit by raising its price above the market-clearing candidate. Intermediate capacities can then produce Edgeworth-type price dispersion in mixed strategies rather than a pure equilibrium. This is why continuation payoffs become harder to compute.

Do not infer that *every* capacity pair outside Equation (4.3) lacks a pure equilibrium. For example, if each firm has capacity at least a, both can serve all demand at zero price and the ordinary zero-cost Bertrand outcome is a pure equilibrium. The no-pure-equilibrium conclusion concerns the relevant intermediate constrained region, not all large capacities.

Proportional rationing

Under proportional rationing, units of the low-price firm are allocated randomly among buyers willing to pay its price. If firm 1 charges p_1 and $a - p_1 > k_1$, the probability of being left unserved there is

$$1 - \frac{k_1}{a - p_1}.$$

For $p_2 > p_1$, firm 2's residual demand becomes

$$D_2^P(p_2; p_1) = \left(1 - \frac{k_1}{a - p_1}\right)(a - p_2). \tag{4.8}$$

The high-price firm can now reach some high-value consumers who happened not to get the cheaper units. Efficient rationing instead served those consumers first.

For fixed p_1, ignoring its own capacity, firm 2 maximizes a positive constant times $p_2(a - p_2)$, with optimal price $a/2$. This is the zero-production-cost monopoly price for the residual *revenue* problem. At a common market-clearing p^*, an upward deviation is unattractive if $p^* \geq a/2$, equivalent to $k_1 + k_2 \leq a/2$. This is more restrictive than the efficient-rationing conditions.

With proportional rationing, intermediate capacities can again lead to mixed pricing, and the first-stage capacity incentives need not reproduce Cournot. The lecture discusses cases in which the resulting outcome is more competitive than Cournot. That comparison is a result for a specified two-stage model, not a universal ranking of all rationing rules or all capacity pairs. In particular, an unconstrained residual optimum at $a/2$ does not mean that every equilibrium price or every capacity-constrained firm must charge $a/2$.

4.2 Differentiated products: Cournot vs. Bertrand

Lecture 2 · source PDF pp. 199–241

Model 10: A common quadratic-utility model for price and quantity games

Preferences, budget and demand

To compare the games while keeping preferences fixed, use quasilinear quadratic utility

$$U = q_0 + a(q_1 + q_2) - \frac{1}{2}\left(bq_1^2 + 2dq_1q_2 + bq_2^2\right), \qquad b > |d|. \tag{4.9}$$

Lecture derivation: From utility to inverse and direct demands

State the consumer problem and use the budget. With income y and numeraire price one,

$$\max_{q_0,q_1,q_2 \geq 0} U(q_0, q_1, q_2) \quad \text{subject to} \quad q_0 + p_1q_1 + p_2q_2 = y.$$

Since utility is increasing in q_0, the budget binds. Substitute $q_0 = y - p_1q_1 - p_2q_2$. The reduced objective is

$$F(q_1, q_2) = y + (a - p_1)q_1 + (a - p_2)q_2 - \frac{b}{2}q_1^2 - dq_1q_2 - \frac{b}{2}q_2^2.$$

Each purchased unit costs one unit of numeraire per dollar; these opportunity costs produce the terms $-p_iq_i$.

Lecture derivation: From utility to inverse and direct demands (continued)

Differentiate the reduced objective, with prices fixed. For an interior solution,

$$\frac{\partial F}{\partial q_1} = a - p_1 - bq_1 - dq_2 = 0,$$
$$\frac{\partial F}{\partial q_2} = a - p_2 - dq_1 - bq_2 = 0.$$

In particular, differentiating $-dq_1q_2$ with respect to q_1 gives $-dq_2$. Rearranging the two first-order conditions gives inverse demands

$$P_1 = a - bq_1 - dq_2, \qquad P_2 = a - dq_1 - bq_2.$$

The Hessian of F is $-\begin{pmatrix} b & d \\ d & b \end{pmatrix}$, with eigenvalues $-(b+d)$ and $-(b-d)$. Both are negative under $b > |d|$, so the reduced objective is strictly concave. This statement concerns (q_1, q_2) after budgeting; raw utility is linear in q_0.

Solve the same equations for quantities. Write $bq_1+dq_2 = a-p_1$ and $dq_1+bq_2 = a-p_2$. Multiply the first by b and subtract d times the second:

$$(b^2 - d^2)q_1 = a(b-d) - bp_1 + dp_2.$$

Eliminating q_1 analogously yields

$$Q_1 = \frac{a(b-d) - bp_1 + dp_2}{b^2 - d^2} = \widetilde{a} - \widetilde{b}p_1 + \widetilde{d}p_2,$$
$$Q_2 = \frac{a(b-d) + dp_1 - bp_2}{b^2 - d^2} = \widetilde{a} + \widetilde{d}p_1 - \widetilde{b}p_2,$$

where $\widetilde{a} = a/(b+d)$, $\widetilde{b} = b/(b^2-d^2)$ and $\widetilde{d} = d/(b^2-d^2)$. Inverse and direct coefficients differ because one system is the inverse of the other.

These linear demands require positive quantities and an affordable positive numeraire. If a calculated quantity is negative, impose the appropriate zero constraint and re-solve the consumer problem; separately truncating both formulas can misstate the other good's demand.

The parameter d/b summarizes the cross-quantity link relative to the own effect. For substitutes, $0 < d/b < 1$; values close to one approach the homogeneous-good limit and values near zero approach independence. For complements, $-1 < d/b < 0$.

A singular endpoint is not a Leontief utility function

At $d = b$, utility depends only on $q_1 + q_2$, giving perfect substitutes. The opposite boundary $d = -b$ makes the quadratic form singular in the common-output direction; with $a > 0$, unconstrained utility along $q_1 = q_2$ is not bounded above net of sufficiently low prices. It is therefore not legitimate to treat the same regular formulas at $d/b = -1$ as a well-behaved perfect-complements model. A Leontief specification requires a different utility representation.

Best responses expose the difference between the games

With constant marginal costs c_i, a Cournot firm maximizes

$$(a - bq_i - dq_j - c_i)q_i.$$

Its interior best response is

$$R_i^C(q_j) = \frac{a - c_i - dq_j}{2b}, \qquad \frac{\partial R_i^C}{\partial q_j} = -\frac{d}{2b}. \tag{4.10}$$

A Bertrand firm instead maximizes

$$(p_i - c_i)(\tilde{a} - \tilde{b}p_i + \tilde{d}p_j).$$

Its interior best response is

$$R_i^B(p_j) = \frac{\tilde{a} + \tilde{b}c_i + \tilde{d}p_j}{2\tilde{b}} = \frac{a(b-d) + bc_i + dp_j}{2b}, \qquad \frac{\partial R_i^B}{\partial p_j} = \frac{d}{2b}. \tag{4.11}$$

The sign reverses because the rival holds a different strategic object fixed.

Demand relationship	Quantity choices	Price choices
Substitutes: $d > 0$	Downward responses; strategic substitutes	Upward responses; strategic complements
Independent: $d = 0$	No strategic interaction	No strategic interaction
Complements: $d < 0$	Upward responses; strategic complements	Downward responses; strategic substitutes

Product complementarity describes how demand responds to another price. Strategic complementarity describes how *marginal profit from an action* responds to a rival's action. They are different concepts.

Explicit symmetric outcomes

Following the lecture, set $c_1 = c_2 = 0$ to keep the comparison transparent. Under Cournot symmetry, $q_i = q_j = q^C$ gives

$$2bq^C + dq^C = a,$$

while the first-order condition implies $p^C = bq^C$. Under Bertrand symmetry, $p_i = p_j = p^B$ gives

$$(2\tilde{b} - \tilde{d})p^B = \tilde{a},$$

and the price first-order condition implies $q^B = \tilde{b}p^B$. Therefore

Same preferences, different commitments

$$q^C = \frac{a}{2b + d}, \qquad p^C = \frac{ab}{2b + d}, \tag{4.12}$$

$$p^B = \frac{a(b-d)}{2b - d}, \qquad q^B = \frac{ab}{(b+d)(2b-d)}. \tag{4.13}$$

These are interior outcomes for $a > 0$ and $b > |d|$, with a positive numeraire balance or the corresponding aggregate-demand interpretation.

Lecture derivation: Compare prices and quantities algebraically

Compare prices directly:

$$p^C - p^B = \frac{ab(2b-d) - a(b-d)(2b+d)}{(2b+d)(2b-d)} \tag{4.14}$$

$$= \frac{ad^2}{4b^2 - d^2} \geq 0. \tag{4.15}$$

The inequality is strict whenever $d \neq 0$; at $d = 0$, the independent firms choose identical monopoly outcomes under either action variable. Similarly,

$$q^B - q^C = \frac{ad^2}{(b+d)(4b^2 - d^2)} \geq 0.$$

Thus the specified symmetric quadratic model gives lower prices and greater output under Bertrand for both substitutes and complements, with equality at independence. This is a model-specific comparison, not a theorem covering every asymmetric demand system.

Lesson 11: the strategic variable changes competitiveness

1. In this symmetric quadratic model, $p^B \leq p^C$ and $q^B \geq q^C$, strictly when $d \neq 0$. At independence the two games coincide.
2. Fixing the rival's price makes perceived demand more responsive than fixing its quantity. This reduces the chosen price–cost margin.
3. The welfare ranking follows here from higher output below the efficient level. Profit rankings still depend on whether products are substitutes or complements.

Why perceived demand is more elastic under price competition

In Cournot competition the rival's quantity is fixed, so the perceived direct relationship is

$$q_i = \frac{a - p_i - dq_j}{b}, \qquad \left|\frac{\partial q_i}{\partial p_i}\right| = \frac{1}{b}.$$

In Bertrand competition the rival's price is fixed, so

$$q_i = \tilde{a} - \tilde{b}p_i + \tilde{d}p_j, \qquad \left|\frac{\partial q_i}{\partial p_i}\right| = \frac{b}{b^2 - d^2}.$$

For $d \neq 0$, $b/(b^2 - d^2) > 1/b$. At the same feasible price-quantity point, fixing the rival's price therefore makes perceived own demand more responsive. A price cut can attract substitution from the rival as well as increase consumption of the category. This sharper volume response reduces the optimal markup.

For substitute products, stronger differentiation means d/b moves toward zero, and the gap between the two games shrinks. For complementary products, weakening the interaction likewise means moving $|d|/b$ toward zero. An arbitrary decrease in a negative d strengthens complementarity rather than moving the goods toward independence.

Welfare and profits need separate comparisons

At a symmetric quantity q and zero costs, total surplus is

$$\mathrm{TS}(q) = 2aq - (b+d)q^2.$$

Its maximum is at $q^{eff} = a/(b+d)$. Both oligopoly outputs are below that level, and $q^B \geq q^C$. Thus the higher Bertrand output raises total surplus in this model. Quasilinear consumer surplus at symmetry is $(b+d)q^2$, so consumers also benefit.

Profit rankings depend on the sign of d:

$$\pi^C = \frac{a^2 b}{(2b+d)^2}, \tag{4.16}$$

$$\pi^B = \frac{a^2 b(b-d)}{(b+d)(2b-d)^2}, \tag{4.17}$$

$$\pi^C - \pi^B = \frac{2a^2 b d^3}{(2b+d)^2(b+d)(2b-d)^2}. \tag{4.18}$$

Firms prefer Cournot if products are substitutes, and Bertrand if they are complements; at independence profits coincide. Lower prices need not imply lower profit when complementarity induces enough additional demand.

The same action label can hide different incentives

Under substitutes, a rival's output expansion makes own expansion less attractive, but a rival's price increase makes own price increases more attractive. Under complements, the feedback reverses. The correct comparison begins with the payoff derivatives, not with whether a market is casually called competitive.

4.3 What is the appropriate modelling choice?

For a monopolist facing a fixed invertible demand curve, price and quantity are two ways to describe one choice. In an oligopoly, changing the strategic variable changes what competitors commit to and therefore changes residual demand.

Under homogeneous price competition, a rival is willing to serve any feasible quantity at its posted price. Firm i's demand is the entire market below that price and zero above it, with sharing at a tie. This is a jump, not an ordinary differentiable demand elasticity at the tie. Under quantity commitment, residual demand is $[Q(p_i) - q_j]_+$, a much smoother constraint.

A price game is more plausible when quantities adjust readily and prices are costly to change, as with a printed catalogue or a posted price list. A quantity or capacity model is more plausible when stock, seats or rooms are committed far ahead of demand while prices remain flexible. Package holidays booked a year before the season and later discounted provide the lecture's illustration. These are guides for choosing a model, not empirical proof of the chosen conduct.

Case 4: print on demand and digital distribution

1. **Technology and costs.** Batch printing uses large runs to lower unit cost, committing inventory in advance. Print on demand makes small runs economical, shifting costs from fixed to variable categories; variable cost per book is usually higher.
2. **Inventory implications.** Print on demand keeps the backlist available, avoids stock-outs or large reprint costs, and saves warehousing. It is especially suitable for low or unpredictable demand.
3. **Which model fits?** With batch printing, prices adjust to sell a precommitted stock. With print on demand, quantity adjusts at a posted price. The technology can change the strategic commitment itself.
4. **Film analogy.** Moving a multi-gigabyte digital film file costs much less than shipping 35mm prints, facilitating small releases and flexible distribution. Digital production and distribution change together; physical screening seats and locations can still constrain supply.

SECTION 5

Strategic substitutes and strategic complements

Lecture 2 · source PDF pp. 242–265; 290–296

A general definition based on marginal incentives

Let firm i choose a scalar action x_i from a compact interval, with profit $\pi_i(x_i, x_{-i}; \theta)$. The action can be price, output, quality, advertising or investment. Rivals' actions are *strategic complements* when increasing a rival action raises the marginal return to increasing one's own action. In a twice-differentiable model, increasing differences are expressed as

$$\frac{\partial^2 \pi_i}{\partial x_i \partial x_j} \geq 0 \qquad (j \neq i). \tag{5.1}$$

Strategic substitutes satisfy the reverse inequality. These properties concern incentives; they are not definitions of substitute or complementary *products*.

Deriving the slope of a reaction function

Lecture derivation: The reaction slope by implicit differentiation

Suppose firm i has an interior response $x_i = R_i(x_j)$ and $\frac{\partial^2 \pi_i}{\partial x_i^2} < 0$. The response is defined by

$$\left.\frac{\partial \pi_i(x_i, x_j)}{\partial x_i}\right|_{x_i = R_i(x_j)} = 0.$$

This equation holds for every nearby rival action, not just at the equilibrium. Differentiating the identity with respect to x_j gives

$$\frac{\partial^2 \pi_i}{\partial x_i^2} \frac{\mathrm{d}R_i(x_j)}{\mathrm{d}x_j} + \frac{\partial^2 \pi_i}{\partial x_i \, \partial x_j} = 0.$$

Both partial derivatives are evaluated at $(R_i(x_j), x_j)$. Isolating the response slope,

$$\frac{\mathrm{d}R_i(x_j)}{\mathrm{d}x_j} = -\frac{\dfrac{\partial^2 \pi_i}{\partial x_i \, \partial x_j}}{\dfrac{\partial^2 \pi_i}{\partial x_i^2}}. \tag{5.2}$$

The own second derivative is negative. Therefore the response has the same sign as the cross-partial: upward for strategic complements and downward for strategic substitutes.

What the first-order proof assumes

The equality applies to differentiable interior responses with a nonzero own second derivative. At a constrained boundary, use the relevant inequality or best-response correspondence. Strict own concavity makes an existing own optimum unique; it does not itself guarantee that a solution to the simultaneous equilibrium equations exists or is unique.

Discrete actions and increasing differences

Suppose $x_i, x_j \in \{0, 1\}$. The incentive to choose 1 rather than 0 is

$$\Delta_i(x_j) = \pi_i(1, x_j) - \pi_i(0, x_j).$$

Complementarity means $\Delta_i(1) \geq \Delta_i(0)$, equivalently

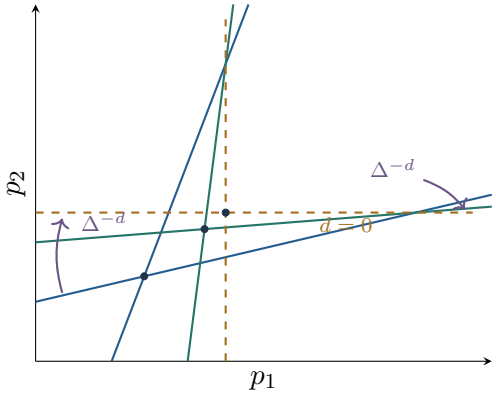
$$\pi_i(1,1) + \pi_i(0,0) \geq \pi_i(1,0) + \pi_i(0,1). \tag{5.3}$$

The diagonal payoff sum exceeds the off-diagonal sum. Reverse the inequality for substitutes. No differentiability is needed, and a best response may jump upward rather than vary smoothly.

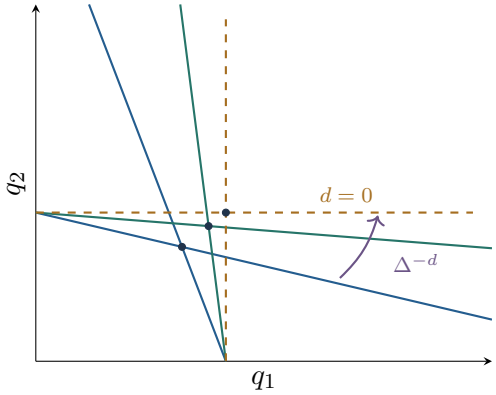
An investment incentive with two actions

Let $\pi_i(x_i, x_j) = Ax_i - Kx_i + Bx_ix_j$. Investing rather than not investing gives $A - K$ when the rival does not invest and $A - K + B$ when it does. If $B > 0$, investment incentives are complementary. If $B < 0$, the rival's investment reduces the return to own investment, making the actions substitutes. This calculation concerns the interaction payoff, regardless of the name of the industry.

Price: strategic complements



Quantity: strategic substitutes



Course figures: reaction slopes and decreasing product interaction. Source PDF pp. 252–253. Each colour represents both interior reaction equations at one parameter: blue $d = 0.6$, green $d = 0.2$, dashed amber $d = 0$, with $a = 10, b = 1, c_i = 0$. The three points are the corresponding intersections. The course symbol Δ^{-d} denotes a decrease in d; the extra lines are response curves under different parameters. At $d = 0$, each action is independent of its rival. In quantities the own-response intercept stays at $a/(2b)$; in prices the intercept also changes. Boundary branches are not shown.

Equilibrium structure and monotone comparative statics

The course identifies three uses for complementarity:

1. **Existence:** appropriate global regularity and lattice conditions support a pure equilibrium.
2. **Order:** smallest and largest equilibria organize a potentially multiple equilibrium set.
3. **Comparative statics:** ordered parameter changes can shift those extremal equilibria predictably, even when no closed form is available.

In the elementary linear examples equilibrium is unique; the third use is the main application. More generally, on compact lattice strategy sets, with the standard continuity/semicontinuity

and increasing-differences conditions for supermodular games, strategic complementarity provides existence and a smallest and largest pure equilibrium. Multiple equilibria may remain. These global results do not require smooth first-order conditions, and complementarity by itself without the accompanying regularity conditions is not a complete existence theorem.

Suppose a parameter θ increases marginal own profit for every firm at every relevant action profile:

$$\frac{\partial^2 \pi_i}{\partial x_i \, \partial \theta} \geq 0. \tag{5.4}$$

Under the standard supermodular-game conditions, both the smallest and largest equilibria weakly rise with θ. If equilibrium is unique, the comparison is unambiguous. The result orders corresponding extremal equilibria; it does not compare arbitrarily selected equilibria from two different equilibrium sets.

Lesson 12: complementarity orders equilibrium responses

1. Assume the parameter raises marginal own profit globally and the supermodular-game regularity conditions hold.
2. The smallest and largest equilibrium action profiles weakly increase with the parameter. With uniqueness, this gives one unambiguous equilibrium comparison.
3. An own response shift holds rivals fixed. The local total equilibrium derivative must additionally solve their feedback system and use its relevant stability condition.

A direct shift is not yet an equilibrium change

Lecture derivation: Separate the direct parameter shift from equilibrium feedback

Write an interior response as $R_i(x_j, \theta)$. Differentiating its first-order condition while holding x_j fixed gives

$$t_i \equiv \frac{\partial R_i}{\partial \theta} = -\frac{\frac{\partial^2 \pi_i}{\partial x_i \, \partial \theta}}{\frac{\partial^2 \pi_i}{\partial x_i^2}} \geq 0. \tag{5.5}$$

This is the direct response shift from the appendix. An equilibrium change must also account for the rival's endogenous movement. For two firms, let $r_i = \frac{\partial R_i}{\partial x_j}$. Then

$$\frac{dx_1^*}{d\theta} = t_1 + r_1 \frac{dx_2^*}{d\theta}, \qquad \frac{dx_2^*}{d\theta} = t_2 + r_2 \frac{dx_1^*}{d\theta}.$$

Solving,

$$\frac{dx_1^*}{d\theta} = \frac{t_1 + r_1 t_2}{1 - r_1 r_2}, \qquad \frac{dx_2^*}{d\theta} = \frac{t_2 + r_2 t_1}{1 - r_1 r_2}. \tag{5.6}$$

For complements, $r_1, r_2 \geq 0$. If $r_1 r_2 < 1$, both derivatives are nonnegative and the feedback amplifies the direct shifts. The condition on the denominator is needed for this particular local differentiable argument. A global lattice-theoretic monotonicity theorem and this local stable-equilibrium calculation are related but different arguments.

A positive feedback loop

$\theta \uparrow \Longrightarrow$ R_1, R_2 shift upward $\Longrightarrow$ $x_1, x_2 \uparrow \Longrightarrow$ further increases in each other's incentives.

Under substitutes, the rival's response can offset the direct effect. Signs then require solving the system rather than observing only the direction of a single response shift.

Application: more differentiation and higher prices

In the quadratic price game with substitutes $0 < d < b$ and zero marginal costs,

$$R_i(p_j; d) = \frac{a(b-d) + dp_j}{2b}.$$

As d falls, the response's intercept rises and its slope falls. At a fixed rival price,

$$\frac{\partial R_i}{\partial d} = \frac{p_j - a}{2b}.$$

For economically relevant prices below a, a decrease in d raises the best response. Both firms' higher prices then reinforce each other.

To reproduce the marginal-profit calculation carefully, write

$$\pi_i^B = \frac{p_i[a(b-d) - bp_i + dp_j]}{b^2 - d^2}.$$

Differentiating twice and only then evaluating at symmetric prices $p_i = p_j = p$ gives

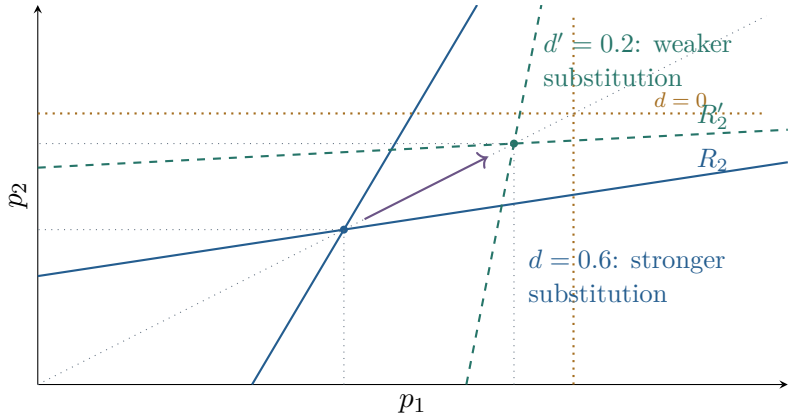
$$\left.\frac{\partial^2 \pi_i^B}{\partial p_i \partial d}\right|_{p_i=p_j=p} = -\frac{(b-d)^2(a-p) + 2bdp}{(b^2-d^2)^2} < 0 \tag{5.7}$$

for $0 \le d < b$ and $0 < p < a$. The factor p in $2bdp$ is essential. A decrease in d raises marginal profit in this local comparison.

The explicit equilibrium offers a separate check:

$$p^B(d) = \frac{a(b-d)}{2b-d}, \qquad \frac{dp^B}{dd} = -\frac{ab}{(2b-d)^2} < 0. \tag{5.8}$$

Thus weaker substitutability raises price in the specified model. A sign evaluated only around symmetric prices is not, by itself, proof of a global increasing-differences condition over the entire strategy space; the explicit formula settles this example directly.



Course figure: differentiation shifts both price responses. Source PDF pp. 261–262. With $a = 10, b = 1, c_i = 0$, decreasing d from 0.6 to 0.2 raises the intercept and reduces the slope of each price response. The equilibrium moves from $(20/7, 20/7)$ to $(40/9, 40/9)$ after both firms adjust. The dotted independence limit $d = 0$ gives $(5, 5)$; the source's Δ^{-d} arrows refer to this parameter change. A primed response here denotes the changed parameter, not a derivative.

For the illustration, $a = 10$, $b = 1$ and costs are zero. Both price responses become flatter and shift upward at the relevant prices as d falls. Their intersection moves from $p_1 = p_2 = 20/7$ to $p_1 = p_2 = 40/9$. The arrow shows the equilibrium movement, after both firms have adjusted.

SECTION 6

Estimating market power*

Lecture 2 · source PDF pp. 266–285

Instructor-starred material. This part connects the theoretical games to identification and estimation.

A conduct parameter that nests benchmark models

Markets can differ in competitiveness because of technology, demand, entry barriers, coordination and the period being observed. The lecture proposes a common framework for a homogeneous-product, symmetric industry. Let

$$p = P(Q, x), \qquad MC = c(Q, w), \tag{6.1}$$

where x contains demand shifters excluded from costs and w contains cost shifters excluded from demand. The function c is the marginal cost entering a representative firm's equilibrium condition, expressed in terms of aggregate Q under symmetry; with firm-level cost functions, this can embed the relation $q_i = Q/n$.

Lecture derivation: Map benchmark first-order conditions into the conduct equation

Introduce the conduct parameter λ through

$$p + \lambda \frac{\partial P(Q, x)}{\partial Q} Q = c(Q, w). \tag{6.2}$$

This is an equilibrium relation, not a competitive supply curve valid under every conduct model. It nests familiar conditions:

Benchmark	λ	Equilibrium condition
Competition or symmetric homogeneous Bertrand	0	$p = c$
Symmetric n-firm Cournot	$1/n$	$p + \frac{\partial P(Q,x)}{\partial Q} Q/n = c$
Monopoly/joint-profit maximization	1	$p + \frac{\partial P(Q,x)}{\partial Q} Q = c$

Using $\eta = -p/(\frac{\partial P(Q,x)}{\partial Q} Q)$, rearrange Equation (6.2):

$$\boxed{L \equiv \frac{p - c(Q, w)}{p} = -\lambda \frac{\frac{\partial P(Q,x)}{\partial Q} Q}{p} = \frac{\lambda}{\eta}.} \tag{6.3}$$

Even if conduct is unchanged, less elastic demand can produce a larger observed margin. Conduct, demand elasticity and the Lerner index are distinct objects.

Interpretation I: conjectural variations

Suppose firm i conjectures that changing q_i causes aggregate rival output Q_{-i} to change at a constant rate

$$\gamma = \frac{\mathrm{d}Q_{-i}}{\mathrm{d}q_i}.$$

It then perceives $dQ/dq_i = 1 + \gamma$ and uses the condition

$$p + (1+\gamma)\frac{\partial P(Q,x)}{\partial Q} q_i = MC_i.$$

Under symmetry $q_i = Q/n$, comparison with Equation (6.2) gives

$$\lambda = \frac{1+\gamma}{n}, \qquad \gamma = n\lambda - 1. \tag{6.4}$$

In a simultaneous static Cournot Nash game, a unilateral deviation holds rivals' quantities fixed, so $\gamma = 0$ and $\lambda = 1/n$. A nonzero conjecture is therefore not automatically consistent with that static Nash game.

A conjectural-variation parameter can instead summarize effects from an explicitly dynamic setting. The lecture sketches a two-period differentiated price game in which firms see their own sales but do not know a demand intercept. A high first-period price reduces one's current quantity but can lead the rival to infer stronger demand and choose a higher future price. The rival's future quantity can then fall as well, giving a positive relation between the two quantities across periods. This mechanism is an interpretation under specified beliefs and learning, not an inference that a static derivative by itself identifies dynamic conduct.

Interpretation II: an index of market power

Alternatively, treat λ as an empirical conduct index without claiming a literal conjecture. Estimate a demand-and-equilibrium system and ask whether a particular benchmark, such as $\lambda = 1/n$, is compatible with the data. Rejecting that hypothesis does not identify a unique alternative game; it can also reveal misspecified demand, costs, heterogeneity or measurement.

The next issue is identification: do different conduct/cost combinations imply distinguishable observable outcomes?

Why observations of price and quantity are not enough

Suppose the observed equilibrium can be expressed as reduced forms

$$Q = g_1(x,w), \qquad p = g_2(x,w).$$

With suitable exogeneity, support and a specified equilibrium selection, the data can identify features of the conditional distribution or conditional mean of (p,Q) given (x,w). That is not yet identification of the structural cost function or λ. Multiple equilibria and unobserved shocks also require an explicit treatment rather than an assertion that a deterministic reduced form is always identified.

For any candidate λ', Equation (6.2) suggests a candidate cost function

$$c'(Q,w) = c(Q,w) + (\lambda' - \lambda)\frac{\partial P(Q,x)}{\partial Q} Q.$$

If the extra term can be absorbed by an admissible cost function without violating the exclusion of x, the observed data cannot distinguish these parameterizations. Identification requires variation and restrictions that prevent that absorption.

A linear-demand nonidentification example

Let demand and marginal cost be

$$p = a + x - bQ, \qquad c(Q, w) = \kappa_0 + \kappa_1 Q + w.$$

Since $\frac{\partial P(Q,x)}{\partial Q} = -b$, the equilibrium condition becomes

$$p = \kappa_0 + w + (\kappa_1 + \lambda b)Q. \tag{6.5}$$

Even if demand identifies b, the supply-side slope reveals only $\kappa_1 + \lambda b$. A change in conduct can be offset by a change in the cost slope. An excluded demand intercept shifter x alone does not separate the two.

If an excluded shifter z rotates demand,

$$p = a + x - (b + z)Q,$$

then the conduct relation becomes

$$p = \kappa_0 + w + (\kappa_1 + \lambda b)Q + \lambda z Q.$$

Variation in the excluded interaction zQ can separate conduct from cost slope, provided exclusion, support, rank and endogeneity conditions hold. Thus linearity itself does not always defeat identification; the available demand shifts and cost restrictions matter.

Constant-elasticity demand

For inverse demand $P(Q, x) = A(x)Q^{-1/\eta}$, with constant η,

$$\frac{\partial P(Q, x)}{\partial Q} Q = -\frac{p}{\eta},$$

so the conduct equation is

$$p\left(1 - \frac{\lambda}{\eta}\right) = c(Q, w).$$

Changing λ can be confounded with changing the level of marginal costs. Without additional restrictions, this creates another identification problem. A flexible cost function is useful for avoiding restrictive assumptions, but additional flexibility can make conduct harder to identify.

Estimation and interpretation

Both p and Q are equilibrium endogenous variables. A demand shock can affect output, and a cost shock can affect price. Ordinary regression coefficients generally do not recover structural demand or cost slopes without valid excluded instruments or another source of identification.

A structural empirical exercise should proceed in a connected order:

1. Specify demand, costs, excluded shifters, shocks and the proposed conduct relation.
2. Establish why the shifters are exogenous and why they move the system enough to separate demand, costs and conduct.
3. Estimate using a method consistent with simultaneity and the available restrictions.
4. Propagate uncertainty through λ and $L = \lambda/\eta$. If a conduct estimate is a ratio of coefficients, a small estimated denominator can make naive standard errors unreliable.

5. Test benchmark restrictions, report uncertainty and assess the sensitivity to alternative demand or cost specifications.

Failure to reject Cournot is not proof that Cournot is the unique data-generating game. A descriptive conduct index is also not automatically a causal effect of concentration or a policy intervention.

Historical empirical illustration in the lecture

The slides report Appelbaum's 1982 estimates as approximately $L = 0.07$ for textiles and $L = 0.65$ for tobacco. These are estimates for the study's data and model, illustrating how inferred margins can differ across industries. They are neither current margins nor values established merely by observing the number of firms. Later course material returns to empirical market-power measurement with richer differentiated-product demand.

The chain from theory to measurement

$$\text{Demand and costs} \longrightarrow \text{conduct} \longrightarrow \text{identification} \longrightarrow \text{estimation}.$$

A theory formula is useful empirically only when its components can be separately identified and measured in the market under study.

Formula reference

Model or object	Result, subject to the conditions in the text
Homogeneous Bertrand	$p_1 = p_2 = c$ with equal constant costs and unrestricted capacity.
Asymmetric Bertrand	Selected $p_1 = \min\{c_2, p_1^M\}$, with efficient tie-breaking.
Private uniform costs	$p(c) = (1 + nc)/(n + 1)$; margin $(1 - c)/(n + 1)$.
Covered Hotelling	$p_i = c + \tau$, $q_i = 1/2$, $\pi_i = \tau/2$.
Salop circle	$p = c + \tau/n$, $q_i = 1/n$, $\pi_i = \tau/n^2$.
Quality advantage Δ	Hotelling $q_1^{NE} = 1/2 + \Delta/(6\tau)$; $q_1^{eff} = 1/2 + \Delta/(2\tau)$.
Heterogeneous Cournot	$q_i = [a - nc_i + \sum_{j \neq i} c_j]/[b(n + 1)]$, if all firms are active.
Symmetric Cournot	$q_i = (a - c)/[b(n + 1)]$, $p = (a + nc)/(n + 1)$.
General Cournot markup	$L_i = s_i/\eta$, $\bar{L} = H/\eta$.
Efficient-rationing price	$p = a - k_1 - k_2$ when $2k_i + k_j \leq a$ for both firms.
Differentiated Cournot, $c = 0$	$q^C = a/(2b + d)$, $p^C = ab/(2b + d)$.
Differentiated Bertrand, $c = 0$	$p^B = a(b - d)/(2b - d)$; $q^B = ab/[(b + d)(2b - d)]$.
Price comparison	$p^C - p^B = ad^2/(4b^2 - d^2) \geq 0$.
Two-firm feedback	$dx_1^*/d\theta = (t_1 + r_1 t_2)/(1 - r_1 r_2)$.
Conduct condition	$p + \lambda Q \frac{\partial P(Q,x)}{\partial Q} = MC$, $L = \lambda/\eta$.
Conjectural variation	$\lambda = (1 + \gamma)/n$; static Cournot has $\gamma = 0$.

Source concordance

The supplied file *2-static imperfect competition_v1.pdf* contains 296 physical PDF pages, including incremental builds and an appendix. The following concordance covers all physical pages. Appendix proofs are integrated into the corresponding arguments rather than detached from the concepts they explain.

PDF pages	Source material	Destination
1–7	Title, market power and strategic interaction	§1
8–10	DVD-by-mail case, actual/potential competition	§1
11–19	Oligopoly theories and previews of lectures 2/3	§1
20–23	Learning objectives and course outline	§1
24–26	Price-competition introduction and outline	§2.1
27–31	Standard Bertrand demand, deviations and responses	§2.1
32–36	Response diagrams, price grid and open-set problem	§2.1
37–38	Bertrand paradox and extension to n firms	§2.1
39–44	Asymmetric costs, tie-breaking and efficiency rents	§2.1
45–49	Private-cost setup and demand	§2.2
50–57	Bayesian strategy, winning probability and ODE	§2.2
58–59	Candidate equilibrium and deviation check	§2.2
60–64	Cost types, number of firms and expected profits	§2.2
65–68	Single realized seller, listings and repeated shocks	§2.2
69–73	Differentiation, branded drugs and fruit case	§2.3
74–80	Hotelling environment, demand and best responses	§2.3
81–84	Differentiation, participation and local monopoly	§2.3
85–88	Aircraft duopoly, barriers and delivery delays	§2.3
89–95	Salop circle, demand boundaries and n	§2.3
96–101	Asymmetric valuations, equilibrium prices/shares	§2.4
102–105	Planner allocation, inefficient shares and policy	§2.4
106–111	Cournot interpretation, auctioneer and outlines	§3.1
112–117	Linear homogeneous model and residual demand	§3.1

PDF pages	Source material	Destination
118–125	General heterogeneous solution and activity	§3.1
126–133	Duopoly solutions and reaction-shift diagram	§3.1
134–139	Symmetric oligopoly, entry count and limit	§3.1
140	Repeated quantity-competition outline	§3.1
141–145	General demand/cost, markup and curvature	§3.2
146–150	Weighted margins and HHI	§3.2
151–158	Homogeneous price/quantity comparison	§4
159–166	Capacity constraints and capacity-pricing cases	§4.1
167–169	Two-stage game, costs and capacity observability	§4.1
170–176	Efficient rationing, justifications and diagram	§4.1
177–178	Capacity upper bound	§4.1
179–187	Price-stage proof, both deviations and restriction	§4.1
188–193	Capacity stage, Cournot equivalence and mixed prices	§4.1
194–198	Proportional rationing and limitations	§4.1
199–207	Unified quadratic model and both choice problems	§4.2
208–214	Reaction slopes and reversal for complements	§4.2
215–218	Symmetric equilibrium price/quantity comparison	§4.2
219–220	Slopes of perceived residual demand	§4.2
221–223	Differentiation limit, welfare and profit comparison	§4.2
224–236	Appropriate model, residual demand and adjustment	§4.3
237–241	Print-on-demand and digital distribution case	§4.3
242–248	General strategic complements and responses	§5
249–253	Discrete actions, substitutes and diagrams	§5
254–258	Existence, extremal equilibria, parameter theorem	§5
259–262	Amplification and differentiation illustration	§5
263–265	Cross-derivative calculation and explicit check	§5
266–271	Empirical conduct framework and benchmarks	§6
272–278	Conjectural variations and dynamic interpretation	§6
279–282	Lerner index, reduced form and identification	§6

PDF pages	Source material	Destination
283–285	Endogeneity, coefficient ratios and historical estimates	§6
286–289	Appendix: full Hotelling welfare derivation	§2.4
290–294	Appendix: reaction slope via implicit differentiation	§5
295–296	Appendix: parameter shifts and equilibrium movement	§5

Legible handwritten contributions are developed in the discussion of tie-breaking, efficiency rents, willingness-to-pay advantages, perceived residual demand, Hotelling versus general price competition, the Salop boundary calculation, Cournot welfare and the cost-curvature exercise, capacity timing, and the distinction between product and strategic substitutability. The additional derivations are stated as calculations with explicit assumptions, not as literal transcriptions of unclear handwriting.

Sources and Further Reading

Primary course source. Hui Shen. *Static Imperfect Competition*. Advanced Industrial Organization lecture slides, September 4, 2026. Supplied annotated file: `2-static imperfect competition_v1.pdf`, 296 physical pages. This is the source for the lecture sequence, models, cases and the three appendix derivations.

Related lecture. Hui Shen. *Firms, Consumers and the Market*, September 4, 2026. The preceding lecture introduces opportunity costs, quasilinear demand, welfare, monopoly and a dominant firm with a competitive fringe. The formulas needed here are repeated where they enter a derivation.

Historical empirical illustration. Elie Appelbaum (1982). "The Estimation of the Degree of Oligopoly Power." *Journal of Econometrics* 19(2–3), 287–299. The approximate textiles and tobacco Lerner indices in Section 6 are the values reported in the supplied lecture. Their magnitudes are not independently re-estimated here.

Capacity commitments. David M. Kreps and José A. Scheinkman (1983). "Quantity Precommitment and Bertrand Competition Yield Cournot Outcomes." *The Bell Journal of Economics* 14(2), 326–337. The elementary restricted-region argument is developed fully in Section 4.1; the general result requires treatment of continuation price games beyond that elementary region.

Classical starting points. Antoine Augustin Cournot (1838), *Recherches sur les principes mathématiques de la théorie des richesses*; Joseph Bertrand (1883), review discussing Cournot's competition model. The strategic variable and adjustment assumptions explain why their benchmark predictions differ.